VTPEH 6270 - Check Point 06

Data Simulation

Ethan Chacko

2026-04-12

Contents

1 Material & Methods

1.1 Data

```
cleandata_food <- data_food %>%
  rename(
    Location          = LocationAbbr,
    AdultDiabetesPrev = Data_Value
  )
cleandata_food <- cleandata_food %>% select(Location, AdultDiabetesPrev)


#Renaming Variables

cleandata_food <- cleandata_food %>% filter(Location %in% c("GA", "NY"))
#I am comparing diabetes prevalence in adults in NY and GA

cleandata_food <- na.omit(cleandata_food) #Omitting NAs

var_table <- data.frame(
  `Variable Name`  = c("Location", "AdultDiabetesPrev"),
  `Variable Type`  = c("Categorical", "Continuous"),
  `Variable Class (R)` = c("character", "numeric"),
```

```r
  `Description`     = c("U.S. State: Georgia (GA) or New York (NY)",
                        "County-level adult diabetes prevalence (%)")
)

kable(var_table,
      caption = "Description of variables used in this analysis.",
      booktabs = TRUE, align = "l")
```

Table 1: Description of variables used in this analysis.

Variable.Name	Variable.Type	Variable.Class..R.	Description
Location	Categorical	character	U.S. State: Georgia (GA) or New York (NY)
AdultDiabetesPrev	Continuous	numeric	County-level adult diabetes prevalence (%)

```r
cleandata_food$Location <- factor(cleandata_food$Location, levels = c("GA", "NY"))#
```

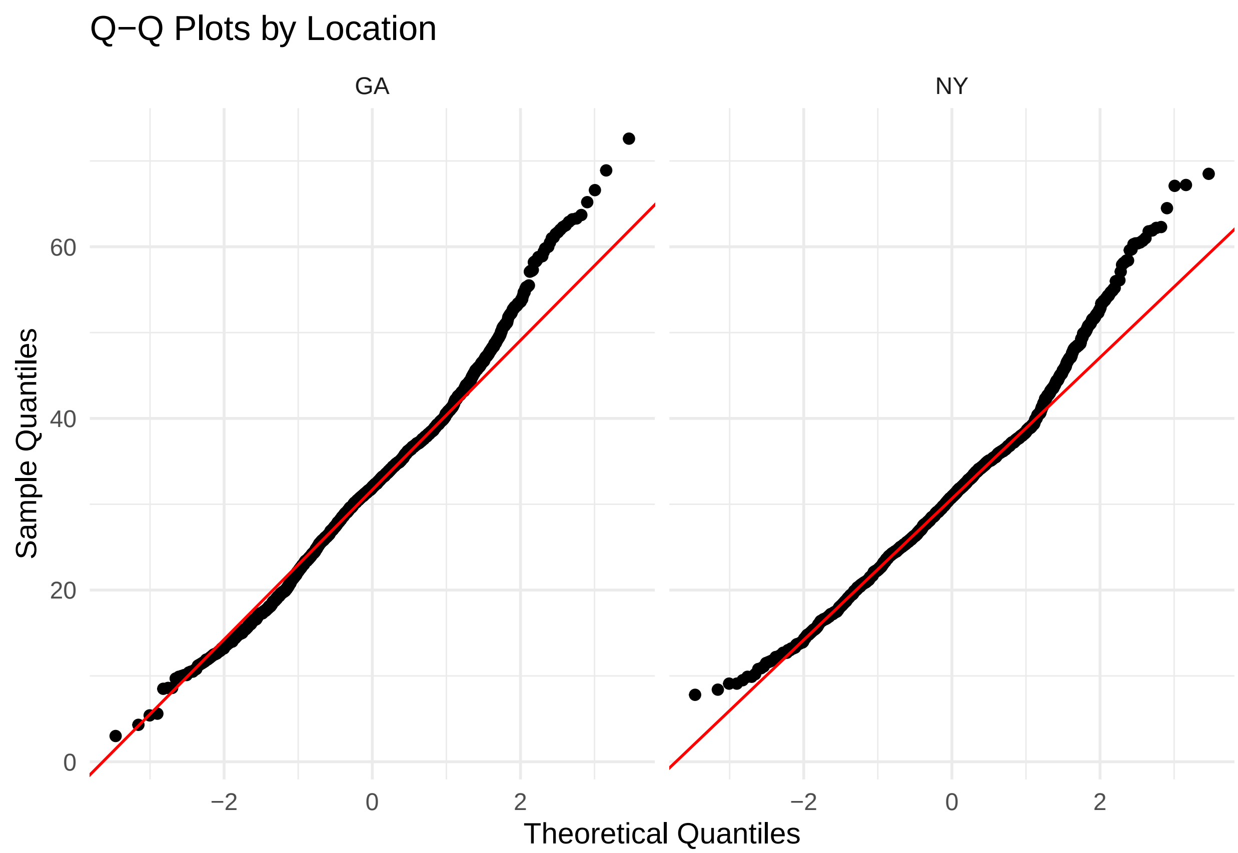
2 Statistical Analyses

```r
t.test(AdultDiabetesPrev ~ Location, data = cleandata_food, var.equal = TRUE)
```

```
Two Sample t-test
```

data: AdultDiabetesPrev by Location t = 2.7156, df = 3777, p-value = 0.006646 alternative hypothesis: true difference in means between group GA and group NY is not equal to 0 95 percent confidence interval: 0.2318118 1.4357310 sample estimates: mean in group GA mean in group NY 31.96389 31.13012

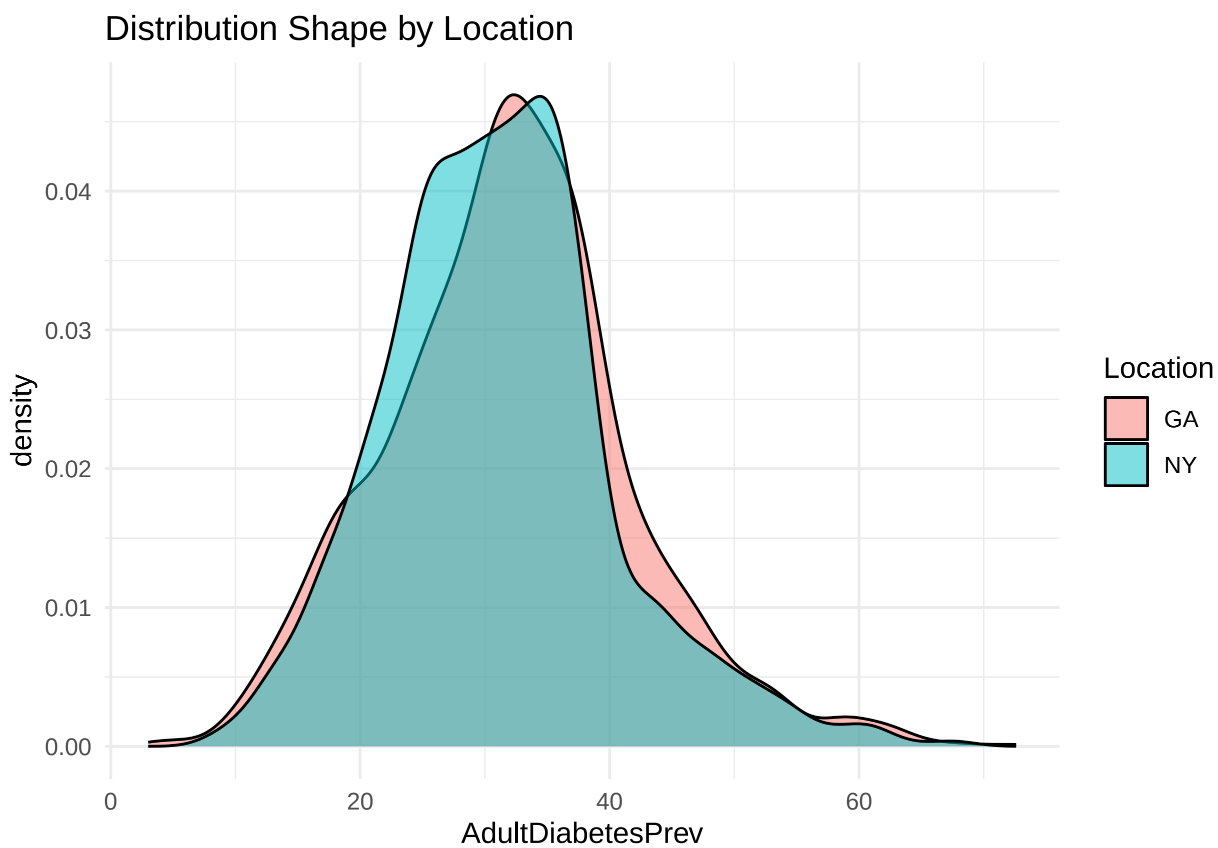
```r
ggplot(cleandata_food, aes(sample = AdultDiabetesPrev)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  facet_wrap(~ Location) +
  labs(title = "Q-Q Plots by Location", x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_minimal()
```




```r
#QQ plot shows normality violated, so we use wilcoxon

#Independent observations can be assumed as one states's diabetes prevalence
#does not affect the next's

ggplot(cleandata_food, aes(x = AdultDiabetesPrev, fill = Location)) +
  geom_density(alpha = 0.5) +
  labs(title = "Distribution Shape by Location") +
  theme_minimal()
```




```
#Similar distribution, can proceed with Wilcoxon

wilcox.test(AdultDiabetesPrev ~ Location, data = cleandata_food, exact = FALSE,
            conf.int = TRUE)
```

```
Wilcoxon rank sum test with continuity correction
```

data: AdultDiabetesPrev by Location W = 1897991, p-value = 0.0007603 alternative hypothesis: true location shift is not equal to 0 95 percent confidence interval: 0.4000322 1.5999368 sample estimates: difference in location 0.9999717

2.1 Code

https://github.com/ethancinthemainframe/R-lemme-cry-about-it-later-.git

3 Results

```
desc_stats <- cleandata_food %>%
  group_by(Location) %>%
  summarise(
    n      = n(),
```

```r
    mean   = round(mean(AdultDiabetesPrev, na.rm = TRUE), 2),
    median = round(median(AdultDiabetesPrev, na.rm = TRUE), 2),
    sd     = round(sd(AdultDiabetesPrev, na.rm = TRUE), 2)
  )
print(desc_stats)
```

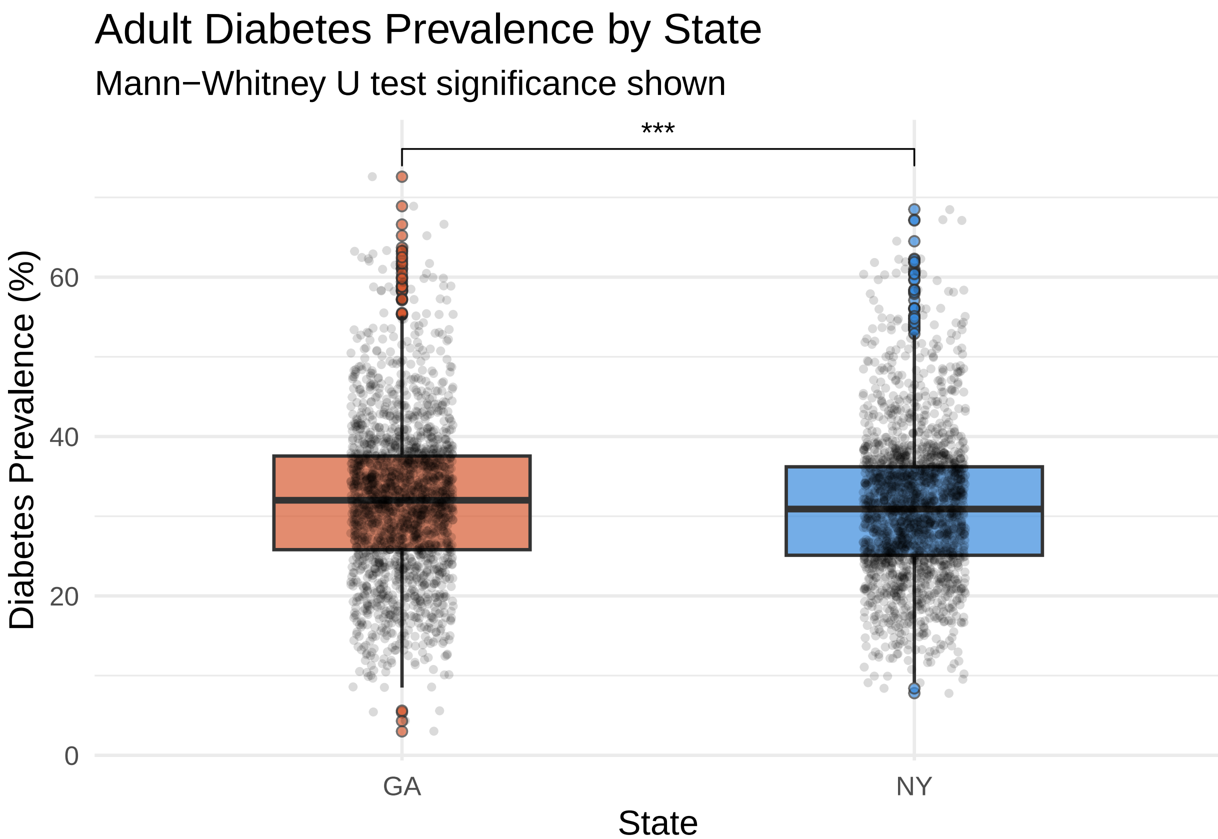
4 A tibble: 2 x 5

Location n mean median sd 1 GA 1883 32.0 32 9.77 2 NY 1896 31.1 30.9 9.09

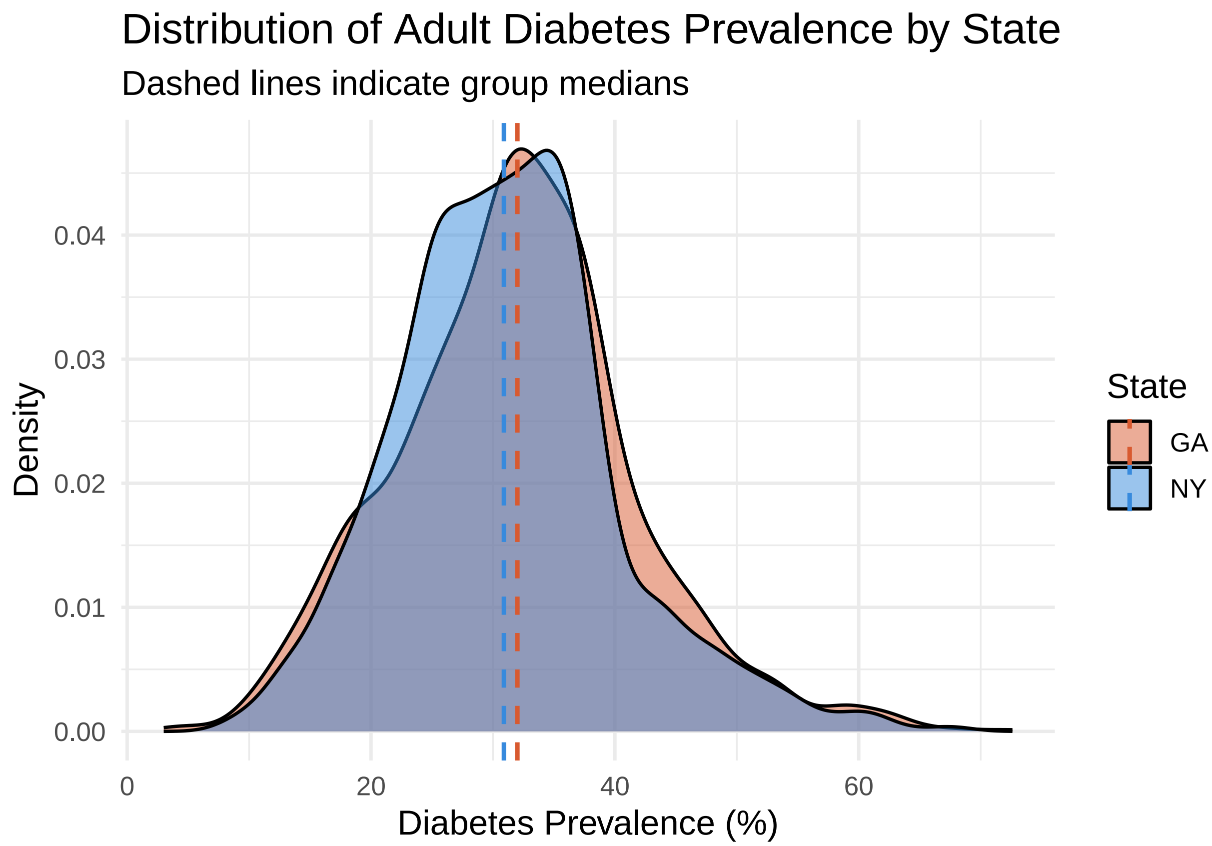
```r
p1 <- ggplot(cleandata_food, aes(x = Location, y = AdultDiabetesPrev, fill = Location)) +
  geom_boxplot(outlier.shape = 21, outlier.size = 1.5, alpha = 0.7, width = 0.5) +
  geom_jitter(width = 0.1, alpha = 0.15, size = 0.8) +
  stat_compare_means(method = "wilcox.test",
                     comparisons = list(c("GA", "NY")),
                     label = "p.signif") +
  scale_fill_manual(values = c("GA" = "#D85A30", "NY" = "#378ADD")) +
  labs(
    title = "Adult Diabetes Prevalence by State",
    subtitle = "Mann-Whitney U test significance shown",
    x = "State",
    y = "Diabetes Prevalence (%)"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "none")

p2 <- ggplot(cleandata_food, aes(x = AdultDiabetesPrev, fill = Location)) +
  geom_density(alpha = 0.5) +
  geom_vline(data = desc_stats,
             aes(xintercept = median, color = Location),
             linetype = "dashed", linewidth = 0.8) +
  scale_fill_manual(values = c("GA" = "#D85A30", "NY" = "#378ADD")) +
  scale_color_manual(values = c("GA" = "#D85A30", "NY" = "#378ADD")) +
  labs(
    title = "Distribution of Adult Diabetes Prevalence by State",
    subtitle = "Dashed lines indicate group medians",
    x = "Diabetes Prevalence (%)",
    y = "Density",
    fill = "State",
    color = "State"
  ) +
  theme_minimal(base_size = 13)

p1
```


Adult Diabetes Prevalence by State
Mann−Whitney U test significance shown
***
Diabetes Prevalence (%)
60
40
20
0
GA
NY
State

```
p2
```



5 AI Use Disclosure Statement

I used Claude AI for every part of this assignment